

Quiz 2: Adversarial and Sequential Decision Making

Weeks 3-5

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Weeks 3-5

Points

20 total

Expected time

35 minutes

Submission

Individual PDF

Instructor verification checklist

- ▶ Work independently.
- ▶ Show backed-up values and arithmetic.
- ▶ State terminal conventions.
- ▶ No electronic communication.

Solution 1.a - Minimax and alpha-beta (2 points)

Question focus. Compute the minimax value and selected branch.

Answer. $\text{MIN}(A)=3$ and $\text{MIN}(B)=2$, so MAX chooses A with value 3.

Solution 1.b - Minimax and alpha-beta (2 points)

Question focus. Search A first. After B's first leaf 2, can B's second leaf be pruned?

Answer. Yes. The root alpha is 3 after A; B's beta becomes 2, so alpha is at least beta and the remaining B child cannot improve MAX's decision.

Solution 1.c - Minimax and alpha-beta (2 points)

Question focus. Does pruning change the backed-up value? Explain.

Answer. No. Alpha-beta removes branches proven irrelevant to the root decision and returns the same minimax value under exact arithmetic.

Solution 2.a - MDP backup (2 points)

Question focus. Write the Bellman action-value expression.

Answer. $Q(s,a)=0.8[0+0.9*5]+0.2[0+0.9*2]$ under the convention that terminal state's listed value is 5.

Solution 2.b - MDP backup (2 points)

Question focus. Compute $Q(s,a)$.

Answer. The result is $0.8*4.5+0.2*1.8=3.6+0.36=3.96$.

Solution 2.c - MDP backup (2 points)

Question focus. Explain what value iteration maximizes.

Answer. For every nonterminal state it computes this expected return for each legal action and sets the new value to the maximum.

Solution 3.a - Q-learning (2 points)

Question focus. Compute the nonterminal target and TD error.

Answer. $\text{Target} = 2 + 0.5 \cdot 3 = 3.5$ and $\text{delta} = 3.5 - 1 = 2.5$.

Solution 3.b - Q-learning (2 points)

Question focus. Compute the updated Q value and state the terminal change.

Answer. New $Q=1+0.25*2.5=1.625$. If terminal, target is reward 2 and new Q is 1.25.

Solution 4.a - Planning versus learning (2 points)

Question focus. What information does each method require?

Answer. Value iteration requires an explicit transition and reward model. Q-learning can learn from sampled transitions without knowing those probabilities.

Solution 4.b - Planning versus learning (2 points)

Question focus. Why must a Q-learning evaluation run use epsilon zero?

Answer. Exploration intentionally chooses suboptimal actions, so leaving epsilon positive confounds learned policy quality with exploration noise.

Edge-case reasoning and expected behavior

- ▶ Return utility without expanding.
- ▶ Reject or normalize only under an explicit data-cleaning policy.
- ▶ Remove the bootstrap term.

Grading rubric

Category	Points	Full-credit evidence
Minimax and alpha-beta	6	Backups and pruning reasoning are correct.
MDP backup	6	Probability, reward, and discount are combined correctly.
Q-learning	4	Target, TD error, and update are correct.
Planning versus learning	4	Model, data, and convergence distinctions are precise.

Reflection exemplar

Answer. Valid reasons include finite exploration, noisy estimates, hyperparameter effects, or a mismatch in transition, reward, or terminal semantics.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.